MIL-C-60303B(MU)
18 February 1971
SUPERSEDING
MIL-C-60303A(MU)
17 November 1969

MILITARY SPECIFICATION

CARTRIDGE, 40MM: RIOT CONTROL,

CS, M674

1. SCOPE

1.1 This specification covers one type of riot control cartridge.

2. APPLICABLE DOCUMENTS

2.1 Government documents. The following documents of the issue in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein:

SPECIFICATIONS

FEDERAL

NN-P-530	-	Plywood, Flat-Panel.
QQ-S-781	-	Steel, Strapping, Flat.
TT-W-572	-	Wood Preservative; Water-Repellent.
PPP-B-601	-	Boxes, Wood, Cleated-Plywood.
PPP-B-636	-	Box, Fiberboard.
PPP-F-320	-	Fiberboard; Corrugated and Solid, Sheet Stock (Container Grade), and Cut Shapes.

MILITARY

MIL-P-116	-	Preservation, Methods of.
MIL-B-131	-	Barrier Material, Water Vaporproof, Flexible.
MIL-B-2427	-	Boxes, Ammunition Packing, Wood, Nailed.
MIL-B-3060	-	Boxes, Small Arms Ammunition: M19A1 and M2A1.
MIL-A-5092	-	Adhesive, Rubber Base, General Purpose.
MIL-B-46506	-	Boxes, Ammunition Packing, Wood, Wirebound.
MIL-C-51029	-	Chemical Agent, CS.

FSC 1310

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STANDARDS

MILITARY

MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes.
MIL-STD-129 - Marking for Shipment and Storage.
MIL-STD-1168 - Lot Numbering of Ammunition.
MIL-STD-1440 - Test Facility for Determining Percent Agent Recovery.

DRAWINGS

US ARMY MUNITIONS COMMAND

EDGEWOOD ARSENAL

DL122-3-101 - Cartridge, 40 mm: Riot Control, CS, M674.

(Copies of specifications, standards, drawings and publications required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

3. REQUIREMENTS

3.1 Materials and components.

3.1.1 Materials. All materials cited on Drawing DL122-3-101 or on the subsidiary drawings shall conform to the specifications listed thereon or to the specific characteristics set forth on the drawings.

3.1.2 Components. All components of the cartridge shall conform to the specifications and drawings listed on Drawing DL122-3-101 and subsidiary drawings.

3.2 Filling and assembly. One first fire mix pellet and one delay mix pellet shall be placed in the sleeve (Drawing B122-3-103) in accordance with Drawing D122-3-101. The pellets shall be consolidated under a dead load pressure of 1500 ± 100 pounds. This procedure shall be repeated for an additional two first-fire pellets. The pellets shall be pressed flush with the flange side of the sleeve. The cartridges shall be filled and assembled in accordance with Drawing D122-3-101. Moisture content of the final mix (when loaded into the munition) shall be a maximum of 0.5 percent by weight (see 4.4.4.1). The CS fill and pellets shall be stored in sealed containers with a suitable desiccant (e.g., silica gel) in the proportion of 0.25 pounds to 100 pounds of fill or pellets, if necessary to be stored prior to filling (see 6.4).

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3.3 Functioning. The cartridge shall perform as follows when tested as specified in 4.4.4.2 (see 4.3.3).

(a) When an attempt to fire is made, the munition shall launch (not misfire.)

(b) When fired at an elevation of 22° ± 2°, the munition shall impact at 80 ± 15 meters from the firing point. Ranges exceeding 95 meters shall not be classified as a defect. The muzzle of the launcher shall be 60 ± 2 inches above ground level. The range is measured in a horizontal plane, from the launcher muzzle to the point of impact.

(c) The rubber body shall not rupture at any part except the port membranes.

(d) The munition shall emit a continuous cloud of smoke under pressure for no less than 10 seconds and for no more than 40 seconds.

(e) The time from firing until the first smoke is emitted under pressure shall be no less than 2 seconds and no more than 7 seconds (see 6.5).

(f) The cartridge shall not leak when tested as specified in 4.4.4.2.

3.4 Agent recovery. The smoke cloud from the cartridge shall contain a minimum of 50 percent of the CS in the loaded cartridge when tested as specified in 4.3.4.

3.5 Bandolier assembly. Four cartridges shall be inserted in the bandolier (Drawing D122-3-116) with the primer ends up.

3.6 Preproduction. Prior to the start of regular production, a pre-production lot of cartridges shall be produced in accordance with this specification for examination and tests (see 4.3).

3.7 Workmanship. The cartridge shall be free from contamination (oil, grease or foreign matter) and damage such as cracks, dents or deformation which would impair its use.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection.

4.1.1 Supplier's responsibility. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise

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specified in the contract or order, the supplier may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to specified requirements.

4.1.2 Objective evidence. The supplier shall provide objective evidence acceptable to the contracting officer that the requirements of 3.1 and section 5 for which specific inspection has not been provided in this specification have been satisfied.

4.2 Classification of inspections.

(a) Preproduction inspection (see 4.3)
(b) Quality conformance inspection (see 4.4)

4.3 Preproduction inspection.

4.3.1 Lot. A preproduction lot of 500 cartridges shall be manufactured using the same methods, materials, equipment and processes as will be used during regular production.

4.3.2 Inspection procedure. Sampling and inspection shall be in accordance with table I.

Table I. Preproduction sampling, inspection and acceptance/rejection criteria

Quantity	Requirement	Test	Accept	Reject
4 ounces	3.2	4.4.4.1	0	1
125 cartridges	3.3	4.4.4.2	*8	9
21 bodies	3.4	4.3.4	0	1
146 cartridges	---	4.4.3.3	3	4

*Any round exhibiting one or more of the defects listed in 4.3.3 shall be considered a reject. No one round will be counted as a reject more than once, regardless of how many defects that one round exhibits.

4.3.3 List of defects.

**(a) Cartridge, 40 mm: Riot Control CS, M674 (Drawing D122-3-101).

1. Misfire (body did not launch)
2. Range incorrect (impact at least 65 meters from firing point)
3. Rupture of rubber body other than port membrane

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4. Failure to emit a continuous smoke cloud (10 to 40 seconds)
5. Delay incorrect (2 to 7 seconds)
6. Leakage

**See 3.3

4.3.4 Agent recovery test. Assembled bodies (consisting of the filled body, sleeve assembly and closure cap) shall be removed from the cartridge and placed in the test facility in accordance with MIL-STD-1440. The bodies shall be positioned horizontally by suitable means in the center of the mixing chamber of the tunnel. The assembly shall be functioned by an electric squib or other suitable means. The cloud shall be sampled with filter samplers from 0 to 10 minutes. During the test the wind speed shall be determined and maintained at a constant speed in the sampling section of the tunnel. The dispersed agent shall be collected and analyzed in accordance with MIL-C-51029. Three assembled bodies are to be burned in the tunnel simultaneously.

4.4 Quality conformance inspection.

4.4.1 Lotting. Each lot shall be identified and controlled in accordance with MIL-STD-1168. Contained in any one interfix number of cartridges, no more than one interfix number from one manufacturer of the following components is permitted.

(a) First fire pellets
(b) Delay pellets

4.4.2 Sampling.

4.4.2.1 For examination. Sampling shall be conducted in accordance with MIL-STD-105.

4.4.2.2 For destructive test. Sampling shall be conducted in accordance with MIL-STD-105, inspection level S-4.

4.4.3 Inspection procedure.

4.4.3.1 For examination. Sample cartridges and the level A preparation for delivery shall be examined in accordance with the classification of defects and MIL-STD-105.

4.4.3.2 For destructive test. Sample cartridges shall be tested in accordance with 4.4.4.2 and table II.

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Table II. Functioning

Defects	MIL-STD-105 AQL percent defective	Test
Misfire (body did not launch)	1.5	4.4.4.2
Dud (failure to ignite filling)		
Leakage	2.5	4.4.4.2
Delay time incorrect	4.0	4.4.4.2
Range incorrect		
Burning time incorrect		
Rubber body ruptured		

4.4.3.3 Classification of defects.

(a) Cartridge, 40 mm: Riot Control CS, M674 (Drawing D122-3-101).

Categories	Defects
Critical:	None defined
Major:	AQL 1.0 percent defective
101	Component missing
102	Component incorrectly assembled
103	Protective finish missing or incorrect
104	Marking missing or incorrect
105	Staking missing
106	Cellulose nitrate lacquer missing
107	Insulating compound missing
108	Wrinkle in tape around firing cap assembly and barrel
109	Workmanship (3.7)

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4.4.3.3 Classification of defects (continued).

(b) Preparation for delivery (section 5).

Categories		Defects
Critical:		None defined
Major:	AQL 4.0 percent defective	
	101	Packaging loose
	102	Packing loose
	103	Container incorrect
	104	Barrier bag material incorrect
	105	Barrier bag ruptured
	106	Marking incorrect
	107	Marking missing or illegible
	108	Strapping loose or missing
	109	Instruction sheet missing
	110	Wood preservative missing

4.4.4 Tests.

4.4.4.1 Moisture content. Moisture content shall be determined by placing the sample of mix in a vacuum desiccator for 24 hours at ambient temperature and less than 380 millimeters of mercury. The desiccator shall contain a desiccant such as Drierite or silica gel. The total loss in weight of the sample shall be considered as moisture in the calculation of moisture content.

4.4.4.2 Functioning. The cartridge shall be submerged in water having a temperature of 0° to 10°F above the temperature of the cartridge, to a depth of 1 to 2 inches over body of cartridge for 35 ± 5 minutes. At the end of the soak period, the cartridge shall be dried by wiping and functioned within one hour. A pedestal mounted (see 3.3(b)) M79 launcher shall be used to function the cartridge.

5. PREPARATION FOR DELIVERY

5.1 Preservation and packaging.

5.1.1 Level A. Two bandoliers containing four cartridges shall be packed method 1A-5 of MIL-P-116, within a M19A1 standard ammunition box conforming to MIL-B-3060 (see 6.3). An instruction sheet, Drawing D122-3-125, shall be wrapped around the folded bandolier prior to placing in the box and the web strapping spirally wound in a single layer around this assembly. The cavities in the cover and bottom of the box shall be filled with fiberboard strips. The cover strips (2 each) shall be

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fabricated from fiberboard conforming to type SF, V3s of PPP-F-320 and shall have a dimension of 9 by 3/4 inches. The bottom strips (2 each) shall be fabricated from fiberboard conforming to type CF, W5c of the same specification with a dimension of 9 by 1 inches. Three fiberboard filler pads, having dimensions of 6 1/2 by 3 3/8 inches and conforming to type CF, W5c of PPP-F-320 shall be placed in each end of M19A1 box, which will assure a tight pack. The cover and bottom fiberboard strips shall be secured in place with adhesive conforming to type II of MIL-A-5092. Only a spot application is necessary.

5.2 Packing, level A. Four M19A1 standard ammunition boxes containing thirty-two cartridges, packaged as specified in 5.1, shall be packed in a wirebound box conforming to type I, class 2, grade A of MIL-B-46506 and shall have inside dimensions of 15 1/4 + 7/32 by 11 + 1/8 by 7 1/4 + 1/4 inches, length, width and depth respectively. The boxes or wood parts shall be preserved by completely immersing in wood preservative conforming to composition A of TT-W-572 for a minimum period of three minutes. Prior to packing the cartridges in the treated boxes, or marking the box, the preservative shall be dry as evidenced by the dryness test in MIL-B-2427. Filler pads of weather-resistant fiberboard conformig to type SF, V3s of PPP-F-320 ahd having approximate dimensions of 10 by 5 by 1/8 inches, length, width and thickness respectively, shall be included, two at each end between the ammunition can and the box end. When space limitations do not permit two spacers, one spacer may be omitted from one or both ends. The wirebound box shall be closed in accordance with the requirements specified in MIL-B-46506. In addition two metal straps, 3/4 inches wide by .023 inches thick, conforming to type I, class B, grade 1 of QQ-S-781 shall be applied girthwise around the shipping container, one at each end as close to the ends as practicable.

5.3 Marking. In addition to any special marking required by the contract or order, the M19A1 ammunition boxes shall be marked in accordance with Drawing D122-3-128. Shipping containers shall be marked in accordance with Drawing D122-3-129 (see 5.2, dryness test).

5.4 Palletization. When specified in the contract or order, shipping containers shall be palletized in accordance with Drawing D122-3-130.

6. NOTES

6.1 Intended use. The riot control cartridge is intended for use by law enforcement agencies in such applications as mob dispersion and the protection of law enforcement personnel.

6.2 Ordering data. Procurement documents should specify the following:

(a) Title, number, and date of specification.
(b) The shipment group size and marking (when required).
(c) Level of packaging and packing required.
(d) Preproduction.

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(1) Time allowed for supplier submission of samples for Government test and evaluation after award of contract.
(2) Name and address of test facility and shipping instructions when testing is performed by the Government.
(3) Time required for the Government to notify the supplier whether or not to proceed with production.

(e) Pertinent identification marking information.
(f) Applicable address marking information.
(g) Shipment priority number, if applicable.
(h) Transportation control number, project code and required delivery date.
(i) Palletization if required.

6.3 M19A1 Ammunition Box. The M19A1 standard ammunition boxes, specification MIL-B-3060, will be Government Furnished Equipment (GFE). If this container is not supplied, the procurement document shall specify that the preservation, packaging and packing shall be as stated in appendix.

6.4 Caution. Because of the hazardous nature of loading the item, suppliers are advised to take all necessary precautions to protect personnel from injury while filling and assembling.

6.5 Delay pellet. If necessary, the length of the delay pellets should be varied in order to meet the delay time requirement. Development experience indicates that a burning time of 3.2 ± 0.5 seconds for the pellet loaded sleeve, when tested separately, will yield a delay time for the assembled cartridge that meets the requirements (3.3(e)).

Custodian:

Army - MU

Preparing activity:

Army - MU(EA)

Project No. 1310-A129

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APPENDIX

PREPARATION FOR DELIVERY

10. SCOPE

10.1 This appendix covers the preservation, packaging and packing for the M674 cartridge (see 6.3).

20. Preservation and packaging, level A. Four cartridges, in their bandolier (see 3.5) shall be preserved and packaged in accordance with method 1A-8 of MIL-P-116. The barrier bag shall conform to class 1 of MIL-B-131. Preservative other than that applied to the cartridge during fabrication is not required. The bandolier with cartridges in place shall be folded to form a square shape (two rows of two each). An instruction sheet, Drawing D122-3-125, shall then be wrapped around the folded bandolier and the web strapping shall then be spirally wound in a single layer around this assembly prior to placing it into the bag. The sealed package shall then be placed in a vertical position primer ends down, within a snug-fitting weather-resistant fiberboard box (W6c) conforming to PPP-B-636. The box shall have inside dimensions of 3 3/4 by 3 3/4 by 9 1/4 inches, length, width and depth respectively (depth being the distance between top and bottom closures.) Filler pads of the same material as the box shall be used as necessary to restrict free movement of the contents. The box shall be closed in accordance with the requirements for weather-resistant, style RSC boxes in the appendix of PPP-B-636. In addition all seams and the manufacturer's joint shall be taped in accordance with the requirements in the appendix of PPP-B-636 for the waterproofing of slotted style boxes.

30. Packing, level A. The cartridges, packaged in accordance with 20., shall be packed primer ends down, in boxes conforming to style A, overseas type of PPP-B-601. Plywood shall conform to group A, type 1, grade 3 or 4 of CS35 of NN-P-530 and shall be free of holes or cracks extending through from face to back. When specified in the contract or order, (for SEA destination) the boxes or wood parts shall be immersed for a minimum of 3 minutes in wood preservative conforming to composition A of TT-W-572. Prior to packing the items in the boxes or marking the box treated with wood preservative, the box shall meet the dryness requirement for preservative as specified in MIL-B-2427. Each box shall contain ten packages (40 cartridges) arranged primer ends down, in two rows of five each. The inside dimensions of the box shall be 20 1/2 by 8 1/4 by 10 1/4 inches in length, width and depth respectively. Weather-resistant fiberboard pads fabricated from material conforming to type SF, V3s of PPP-F-320 shall be used as required to assure a tight pack. One instruction sheet (see 20.) shall be placed inside on top of packaged cartridges in each box. The box shall be closed and strapped in accordance with the appendix of PPP-B-601. Strapping shall be type I or IV, class B of QQ-S-781.

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40. Marking. Unless otherwise specified in the contract or order, unit packages shall be marked in accordance with MIL-STD-129. Intermediate packages and shipping containers shall be marked in accordance with Drawing D122-3-127 (see dryness test, 30.).

50. Palletization. Unless otherwise specified in the contract or order, cartridges packed in accordance with 30. shall be palletized in accordance with Drawing D122-3-124.

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SPECIFICATION ANALYSIS SHEET

Form Approved
Budget Bureau No. 22-R255

INSTRUCTIONS: This sheet is to be filled out by personnel, either Government or contractor, involved in the use of the specification in procurement of products for ultimate use by the Department of Defense. This sheet is provided for obtaining information on the use of this specification which will insure that suitable products can be procured with a minimum amount of delay and at the least cost. Comments and the return of this form will be appreciated. Fold on lines on reverse side, staple in corner, and send to preparing activity. Comments and suggestions submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or serve to amend contractual requirements.

SPECIFICATION

ORGANIZATION

CITY AND STATE

CONTRACT NUMBER

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE SPECIFICATION CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A. GIVE PARAGRAPH NUMBER AND WORDING.

B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2. COMMENTS ON ANY SPECIFICATION REQUIREMENT CONSIDERED TOO RIGID

3. IS THE SPECIFICATION RESTRICTIVE?

☐ YES ☐ NO *(If "yes", in what way?)*

4. REMARKS *(Attach any pertinent data which may be of use in improving this specification. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity)*

SUBMITTED BY *(Printed or typed name and activity - Optional)*

DATE

DD FORM 1 JAN 66 1426 REPLACES EDITION OF 1 OCT 64 WHICH MAY BE USED.

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